

Database Replication

Master: (Perform all activities as Root/ Admin)

Step 1 : Create a Database.

Step 2: Modify config files:

Ubuntu : (As root user)

Open terminal

Open config file using nano : `sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf`

Modify the following fields:

`bind-address : 0.0.0.0`

`Server-id : 1` -- uncomment it in the conf file.

Uncomment `log_bin` field as well

`binlog_do_db : <your database name >` that you created in step 1.

Save the file.

Windows:

Go to C:\ ProgramData\ MySQL \ MySQL Server 8.0 \ my.ini Open using Notepad as an Administrator

Modify the following fields:

bind-address : 0.0.0.0

server-id : 1 -- uncomment it in the conf file.

Uncomment log_bin field as well

binlog_do_db : <your database name > that you created in step 1.

Save the file.

Step 3: Restart mysql service: (as root user)

Ubuntu : In Terminal : service mysql restart

Windows : Open cmd as Admin :

net stop <service name>

net start <service name>

You can find service name using : sc query | findstr /I "mysql"

Step 4: Grant replication access to slave

Open mysql :

```
sudo mysql -u root -p
```

Create user '<username>'@'%' identified with
mysql_native_password by '<password>';

Grant replication slave on *.* to '<username>'@'%';

Flush privileges;

```
show master status;
```

You will see a table, keep a note of the file name, position and database name. Will be used in the slave side.

Slave:

Step1: Create Same Database as Master Server.

Step 2: Configure MySQL configuration file.

- Check for connection to master server and vice versa (very crucial step)
 - o ping <master ip>

If it does not work then try turning off firewall

Ubuntu: sudo ufw disable

Windows: netsh advfirewall set privateprofile state off

Ubuntu: (as root user)

- Change mysql configuration file
 - o sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf

Edit the details given below (don't forget to uncomment) :

```
server-id = 2 #change server id to 2
log_bin = /var/log/mysql/mysql-bin.log
binlog_expire_logs_seconds = 2592000
max_binlog_size = 100M
binlog_do_db = mydatabase #database name created
earlier as master
```

relay_log = /var/log/mysql/mysql-relay-bin.log

Windows:

Go to C:\ ProgramData\ MySQL \ MySQL Server 8.0 \ my.ini Open using Notepad as an Administrator

#Add Details given below
server-id = 2
binlog_do_db = mydatabase
relay_log = relay-bin-logs

Step 3: Restart mysql service: (as root user)

Ubuntu : In Terminal : service mysql restart

Windows : Open cmd as Admin :

net stop <service name>

net start <service name>

Step 4: Now in terminal open MySQL command line client

o sudo mysql -u root -p

- In MySQL command line client execute given commands:
 - o stop replica;

- o Reset replica all;
- o CHANGE REPLICATION SOURCE TO
SOURCE_HOST='<master server ip>',
SOURCE_USER='<master server user>',
SOURCE_PASSWORD=' <master server password> ',
SOURCE_LOG_FILE=' <master server log file> ',
SOURCE_LOG_POS= <master server POS number>;
- Start replica;
- Show replica status; -- To see if replica is connected
-

After successful completion of connection with master server and replica creation you will see

“Waiting for source to send event” in replica status.

Now any changes made in Master server it will show in Replica.

Table Level Filtering:

Most of the configuration is same as before. For the table level filtering during replication we need to change configuration file in replica server.

Master Server configuration will be same as before.

Step 1: Change Configuration file

Open configuration file for replica server.

Ubuntu: sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf

Windows: Go to C:\ ProgramData\ MySQL \ MySQL Server 8.0 \ my.ini Open using Notepad as an Administrator

Now add to replicate specific table you need to mention that table in configuration file.

Make sure that you have the same table already created on master and slave database.

Add this in configuration file:

replicate-do-table = database_name.table_name

Step 2:

Restart mysql service: (as root user)

Ubuntu : In Terminal : service mysql restart

Windows : Open cmd as Admin :

net stop <service name>

net start <service name>

You can find service name using : sc query | findstr /l "mysql"

Step 3: On Master server:

show master status;

Get the File name and Positions from the table. Use this information in step 4.

Step 4: Now in terminal open MySQL command line client

- o `sudo mysql -u root -p`
- In MySQL command line client execute given commands:
 - o `stop replica;`
 - o `Reset replica all;`
 - o `CHANGE REPLICATION SOURCE TO`
`SOURCE_HOST='<master server ip>'`,
`SOURCE_USER='<master server user>'`,
`SOURCE_PASSWORD=' <master server password> '`,
`SOURCE_LOG_FILE=' <master server log file> '`,
`SOURCE_LOG_POS= <master server POS number>;`
- `Start replica;`
- `Show replica status; -- To see if replica is connected`
-

After successful completion of connection with master server and replica creation you will see

“Waiting for source to send event” in replica status.

Now you whatever changes are made in table mentioned in configuration file by master server will be replicated into replica server other tables will not be affected.